Math 526, S12 Name: Answer

Quiz 7 Section: SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Reduce $[\mathbf{A}\ \ \mathbf{b}]$ to $[\mathbf{R}\ \ \mathbf{d}]$ (the reduced echelon form) to find the condition on b_1, b_2, b_3 such that $\mathbf{Ax} = \mathbf{b}$ is solvable.

Solution: $[A\ b] = \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 2 & 6 & 4 & 8 & b_2 \\ -2 & -6 & 0 & 0 & b_3 \end{array}\right]$

$$\longrightarrow \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2-2b_1 \\ 0 & 0 & 2 & 4 & b_3+2b_1 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2-2b_1 \\ 0 & 0 & 0 & 0 & b_3-b_2+4b_1 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 1 & 2 & \frac{1}{2}b_2-b_1 \\ 0 & 0 & 0 & 0 & b_3-b_2+4b_1 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 2b_1-\frac{1}{2}b_2 \\ 0 & 0 & 1 & 2 & \frac{1}{2}b_2-b_1 \\ 0 & 0 & 0 & 0 & b_3-b_2+4b_1 \end{array}\right] = [R\ d]$$

$AX=b$ is solvable $\Leftrightarrow b_3-b_2+4b_1=0$

(b) Find the complete solution to $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$.

$$[R\ d] = \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \rightarrow r = 2$$

i) Find a particular solution X_p such that $RX_p = d$.

setting $X_2 = X_4 = 0$, we get

$$\begin{cases} X_1 = \frac{1}{2} \\ X_3 = \frac{1}{2} \end{cases}$$

$$\Rightarrow X_p = (\tfrac{1}{2}, 0, \tfrac{1}{2}, 0)$$

ii) Solve $RX = 0$ to find all special solutions (two).

Let $X_2 = 1$, $X_4 = 0$, we get

$$\begin{cases} X_1 + 3 \cdot 1 = 0 \\ X_3 + 2 \cdot 0 = 0 \end{cases} \Rightarrow X_1 = -3,\ X_3 = 0$$

$$\Rightarrow S_1 = (-3, 1, 0, 0)$$

Let $X_2 = 0$, $X_4 = 1$, We get

$$\begin{cases} X_1 + 3 \cdot 0 = 0 \\ X_3 + 2 \cdot 1 = 0 \end{cases} \Rightarrow X_1 = 0.\ X_3 = -2$$

$$\Rightarrow S_2 = (0, 0, -2, 1)$$

Thus the complete solution is given by

$$X = X_p + C_1 S_1 + C_2 S_2$$

$$= \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} - 3C_1 \\ C_1 \\ \frac{1}{2} - 2C_2 \\ C_2 \end{bmatrix}$$